Abhash Shrestha

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Summary

AI researcher working at the intersection of **Large Language Models**, **Knowledge Graphs**, and **Trustworthy AI**, with a focus on fairness, retrieval-augmented reasoning, and agentic AI systems. First author on all current research outputs, including a manuscript on structural concentration in AI bias research currently under review at *Science Advances* (arXiv preprint available) and a book chapter on knowledge-graph-based LLM bias mitigation currently in production with Elsevier. Designs and empirically evaluates ML/LLM systems combining symbolic knowledge structures with neural methods, and maintains public research artifacts, including an interactive bias-research atlas and open-source repositories.

Research Interests

Large Language Models · Knowledge Graphs · Trustworthy & Fair AI · Agentic AI · Retrieval-Augmented Generation · Neuro-Symbolic AI · Explainable AI · Machine Learning · Natural Language Processing

Education

University of Southampton — Southampton, United Kingdom
MSc - Computer Science ***(Merit, GPA: 3.3–3.5/4.0)*** — *2023 – 2024*

SRM University — Chennai, India
Bachelor of Engineering in Computer Science and Engineering ***(Grade: 84%)*** — *2016 – 2020*

St. Xavier's College — Kathmandu, Nepal
GCE A-Levels — *2013 – 2015*

Publications

Accepted Book Chapters

- **Shrestha, A.**, Chhetri, T. R. **"Integrating Knowledge Graphs and Large Language Models for Bias Mitigation."**
  - **Venue:** Chapter in *Knowledge Graphs and Large Language Models: Current Approaches, Challenges, and Future Directions*, Elsevier (`ISBN 978-0-443-45244-4`)
  - **Contribution:** First author
  - **Status:** In production

Journal Submissions (Preprint Available)

- **Shrestha, A.**, Gautam, S., Sapkota, A., Tiwari, S., Chhetri, T. R. **"Whose Fairness? Structural Concentration in AI Bias Research."**
  - **Venue:** *ACM FAccT 2027*
  - **Contribution:** First author
  - **Status:** Under review
  - **Preprint:** `arXiv:2607.05574`

Manuscripts in Preparation

- **Shrestha, A.**, Chhetri, T. R., Yumusak, S. **"An Analysis of GDPR Tools and Projects for Data Sharing: Challenges and Impact."**
  - **Contribution:** First author
  - **Status:** In preparation; journal submission planned
- **Shrestha, A.**, Chhetri, T. R., Gautam, S. **Quantitative study of AI technology acceptance and AI literacy (Technology Acceptance Model, $N = 468$).**
  - **Contribution:** First author
  - **Status:** In preparation; journal submission planned

Research Experience

CAIR-Nepal: Center for Artificial Intelligence Research — Kathmandu, Nepal
Researcher — *Feb 2024 – Present*

Knowledge-Graph-Augmented Bias Mitigation in LLMs | *GPT, RAG, RDF, SPARQL*

- Proposed and evaluated a hybrid **KG-RAG** framework that integrates structured knowledge graphs into retrieval-augmented generation to steer LLM outputs away from social stereotypes.
- Quantified fairness gains using stereotype scores and demographic fairness metrics; findings to appear as an Elsevier book chapter, currently in production (see Publications).

- First author; co-author: Dr. Tek Raj Chhetri (MIT). Code: `GitHub`.

Structural Concentration in AI Bias Research | *Bibliometrics, Semantic Clustering*

- Led a bibliometric and semantic-clustering study of **692 publications** across five thematic domains, showing that AI fairness research is geographically and institutionally concentrated, with citation impact dominated by a small set of high-income-country institutions.
- First-author manuscript submitted to *Science Advances*; preprint on arXiv (see Publications). Co-authors: Subigya Gautam, Anu Sapkota, Prof. Dr. Sanju Tiwari (Sharda University, TIB Hanover), Dr. Tek Raj Chhetri (MIT).

Bias Research Survey Dashboard | *React, Next.js, TypeScript, CSV, Data Visualization*

- Built and deployed **BiasAtlas**, an interactive React + Next.js dashboard (dynamic visualizations, collapsible tables, CSV-based export) for exploring analytics and trends across the AI bias-mitigation research landscape.
- **Analytical Dashboard**: `View`
- **Repository**: `GitHub`
- **Dataset**: `Hugging Face`

GDPR-Compliant Data Sharing: Tools and Technologies | *Survey, Statistical Analysis*

- First-authored a structured analysis of **31 EU-funded GDPR-related projects** and **15 compliance tools**, revealing a persistent gap between EU-funded GDPR research and sustained real-world tool adoption (manuscript in preparation; see Publications).
- Co-authors: Dr. Tek Raj Chhetri (MIT), Dr. Semih Yumusak (University of Southampton).

AI Technology Acceptance and Literacy Study | *Technology Acceptance Model, Quantitative Methods*

- Designed and conducted a quantitative study ($N = 468$) of determinants of AI technology acceptance, finding evidence that AI literacy mediates the relationship between digital competence and behavioral intent to adopt AI (90% hypothesis support across mediation and reliability analyses).
- First author of a TAM-based analysis of AI acceptance in Nepal (manuscript in preparation; see Publications).
- Co-authors: Dr. Tek Raj Chhetri (MIT), Subigya Gautam (MSc, University of Louisiana at Lafayette).

MSc Dissertation – Shapelet Classifier Optimization | *Python, tsml-eval, Time Series Analysis*

- Benchmarked five statistical quality measures (KS, Wasserstein, F-statistic, Kruskal-Wallis, Mood's Median) for shapelet-based time-series classification, addressing a computational bottleneck relevant to real-time applications in healthcare (ECG/EEG), finance, and IoT.
- Provided empirical guidance on optimal quality-measure selection across dataset characteristics, improving discriminative-pattern discovery while preserving model interpretability.
- Contribution integrated into `tsml-eval`; pending contribution to `AEON`. Advisor: Prof. Dr. Anthony Bagnall (University of Southampton).
- **Thesis**: `DOI:10.5281/zenodo.21253246`

Industrial Experience

Inductiv AI Labs — Singapore

AI Engineer - Research and Development — *Jan 2026 – April 2026*

Stack | *FastAPI, Pydantic AI, LangChain, Deepeval, Jena, RDF, RDFlib, ChromaDB, Sentence-Transformers, Docker*

- Designed an **agentic AI orchestration layer** coordinating RAG retrieval, intent classification, ontology-based reasoning, and human handoff for multi-step task execution.
- Developed a hybrid retrieval pipeline (dense embeddings + BM25 re-ranking) with relevance and quality scoring, and evaluated it using **Deepeval** on answer faithfulness, grounding, and latency.
- Integrated a domain ontology and RDF knowledge graph (Jena, RDFlib) and designed a 3-step **NL-to-SPARQL** pipeline (entity retrieval → ontology pruning → LLM-based query generation with iterative revision) for explainable query answering.
- Implemented multilingual support (Burmese/English, extensible to 50+ languages) using multilingual sentence-transformer embeddings, with provider failover for reliability.
- Built observability tooling (structured logging, request tracing, error analytics) to monitor retrieval quality and system reliability in production.

Earlier Software Engineering Experience — Kathmandu, Nepal

Fintech Solutions (2022 – 2023), Vanilla Transtechnor (2021 – 2022), Smartech Solutions (2020 – 2021) — *2020 – 2023*

- Developed and optimized backend APIs and full-stack web applications (C#, .NET Core, ASP.NET, MS SQL) across fintech and engineering-software domains; contributed to a desktop CAD tool for civil-engineering design and simulation.

Entrepreneurial Experience

Binary Digits Pvt. Ltd. — Kathmandu, Nepal

Co-founder & CTO — *Oct 2024 – May 2025*

- Co-founded the company and led technical strategy, engineering execution, and team development, including hiring, mentoring, and client-facing delivery.

Coursework & Independent Projects

GPT-2 Instruction Tuning with LoRA | *GPT-2, LoRA, Fine-tuning, Hugging Face*

- Comparative experimental study of full fine-tuning vs. LoRA adaptation on GPT-2 (124M) for instruction tuning. Full fine-tuning exhibited catastrophic forgetting and overfitting on 15K examples, while LoRA (0.65% of parameters) preserved base language capability while adding instruction-following behavior.
- **Repository:** GitHub.

Deep Learning Reproducibility (COMP6258) | *PyTorch, CNNs*

- Reproduced experiments on shape vs. texture bias in convolutional neural networks using PyTorch and validated findings from original research papers.
- **Repository:** GitHub.

Computational Finance Modelling & Kalman Filtering | *Python, NumPy, Pandas*

- Applied ARIMA models, portfolio optimization, and pairs-trading strategies to historical FTSE 100 data for market trend analysis; separately built Kalman filters for multi-sensor fusion with mutual-information-based sensor selection.
- **Repository:** GitHub.

Technical Skills

- **AI & ML:** Large Language Models, Deep Learning, Machine Learning, Retrieval-Augmented Generation, Agentic AI Systems, Prompt Engineering, RAG Evaluation (Deepeval), Fairness-Aware AI, Computer Vision, Multilingual NLP
- **Knowledge Graphs:** RDF, OWL, SPARQL, Ontology Engineering, LinkML, Jena, RDFlib
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn, LangChain, FastAPI, Sentence-Transformers, Pydantic
- **Languages:** Python, C#, Java, C++, JavaScript/TypeScript
- **Data & Statistics:** Statistical Testing, Time Series Analysis, Data Analysis & Visualization (NumPy, Pandas, Matplotlib), Vector Databases (ChromaDB)
- **Infrastructure:** Docker, CI/CD, Git

Open Science & Research Outputs

- **BiasAtlas** – public interactive research atlas of AI bias/fairness literature (692 papers, five thematic domains): biasatlas.cair-nepal.org
- **Dataset** – AI Bias Research Landscape (692 publications, bibliographic and citation metadata): Hugging Face
- **Preprint** – "Whose Fairness? Structural Concentration in AI Bias Research": arXiv:2607.05574
- **SABRE-KG** – code for the KG-RAG bias-mitigation pipeline: GitHub
- **Bias Research Dashboard** – source repository: GitHub
- **Portfolio & Code** – Portfolio | GitHub